

2. Combinational Logic

Computer Architecture

Executive Summary

- **Motivation:** We need a higher level of abstraction for larger designs based on theoretical principles to define its functionality.
- **Problem:** Processor operation is complex and we need to define functionality.
- **Overview:**
 - Review of Boolean equations and algebra.
 - Recall definition of axioms and theorems
 - Define the relation between logic and gates (to circuits).
 - Simplification and introduction to Karnaugh maps.
- **Conclusion:** Combinational logic allows to define logic circuits as a building block for a larger design.

Outline

Introduction

Definitions, axioms and theorems

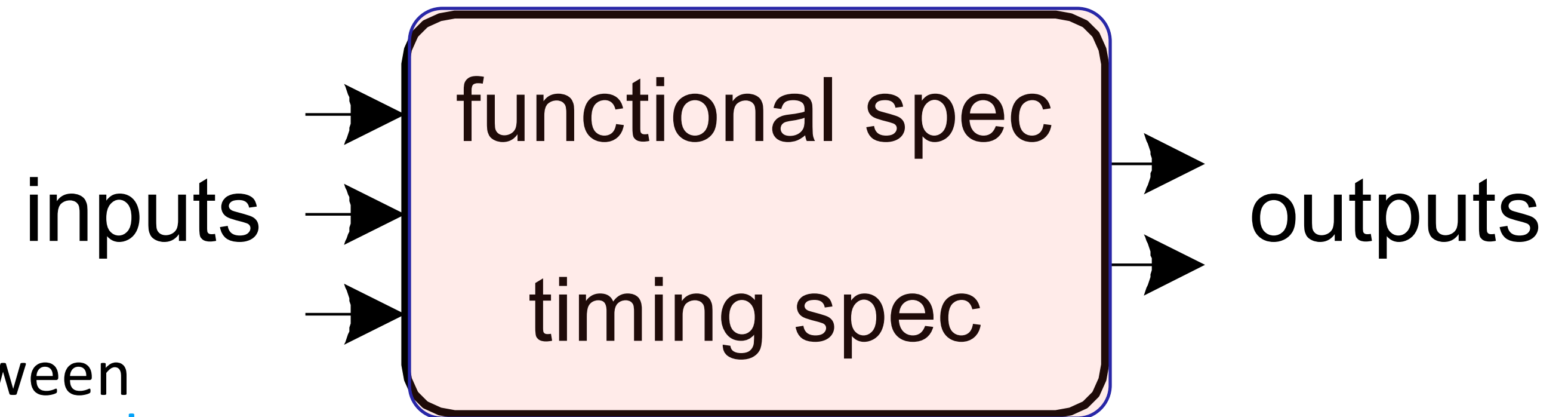
Canonical Forms

Karnaugh Maps

Conclusion

Introduction

- **Elements** of a **logic circuit**:
 - **Inputs**
 - **Outputs**
 - **Functional specification**: defines relation between inputs and outputs.
 - **Timing specification**: defines the delay between inputs (values changing) and outputs (response to change).



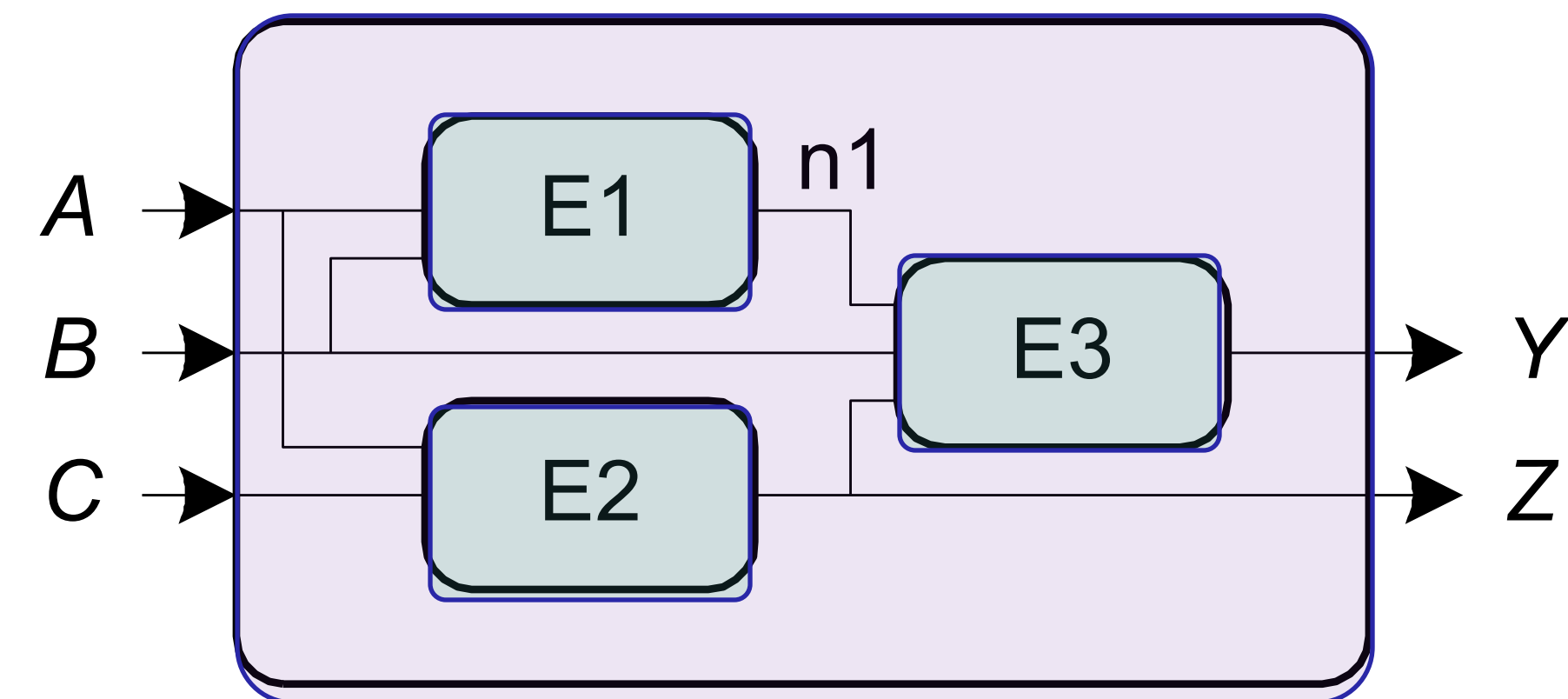
- Elements of a **composed logic circuit**:

- **Nodes**

- Inputs: A, B, C
- Outputs: Y, Z
- Internal: n1

- **(Sub)Circuit elements**

- E1, E2, E3
- Each is also a circuit



Logic Circuits

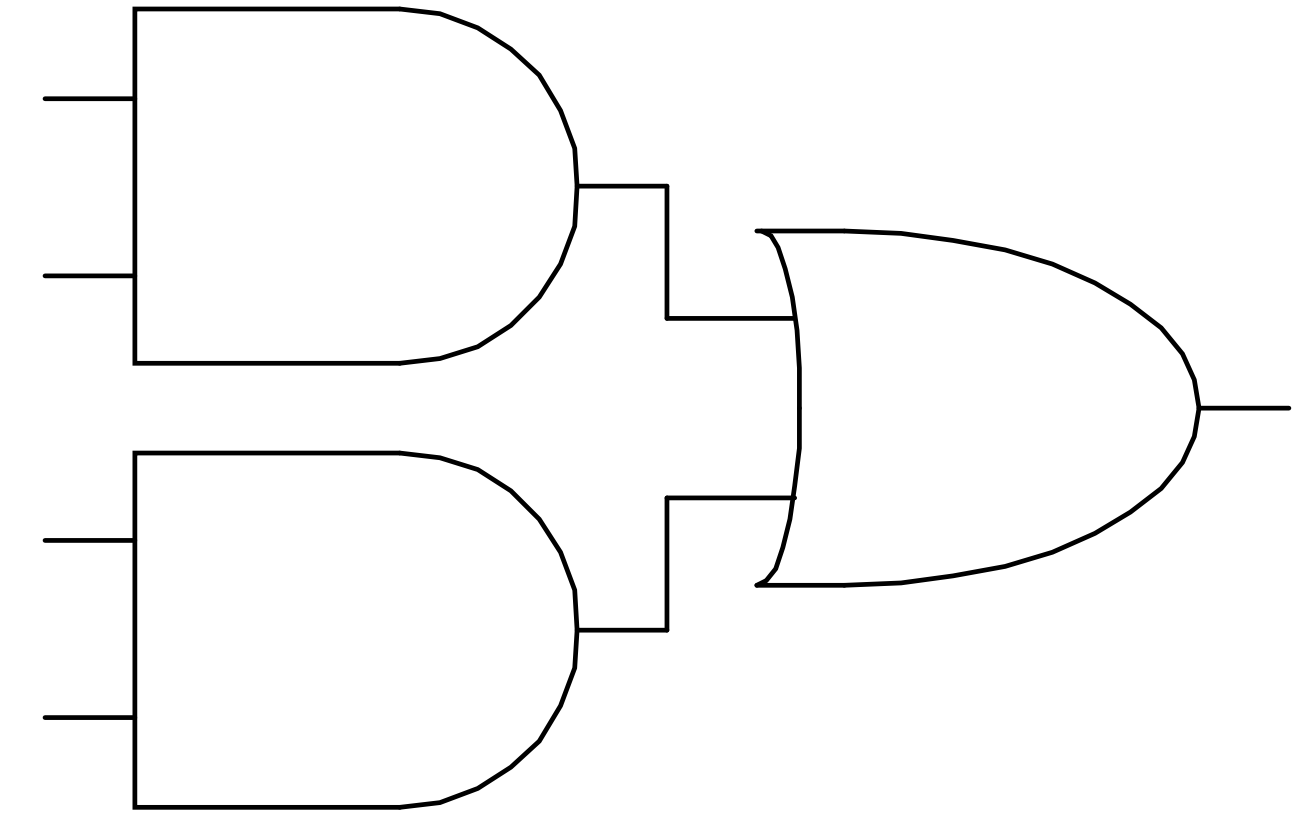
- **Two types** of logic circuits:

1. **Combinational Logic**

- Memoryless
- **Outputs** determined by **combination of current (now) values** of inputs.
- In a **composed combinational circuit**:
 - **Every element** is combinational
 - Every **node** is either an input or connects to *exactly one* output
 - There are **not cyclic paths**

2. **Sequential Logic**

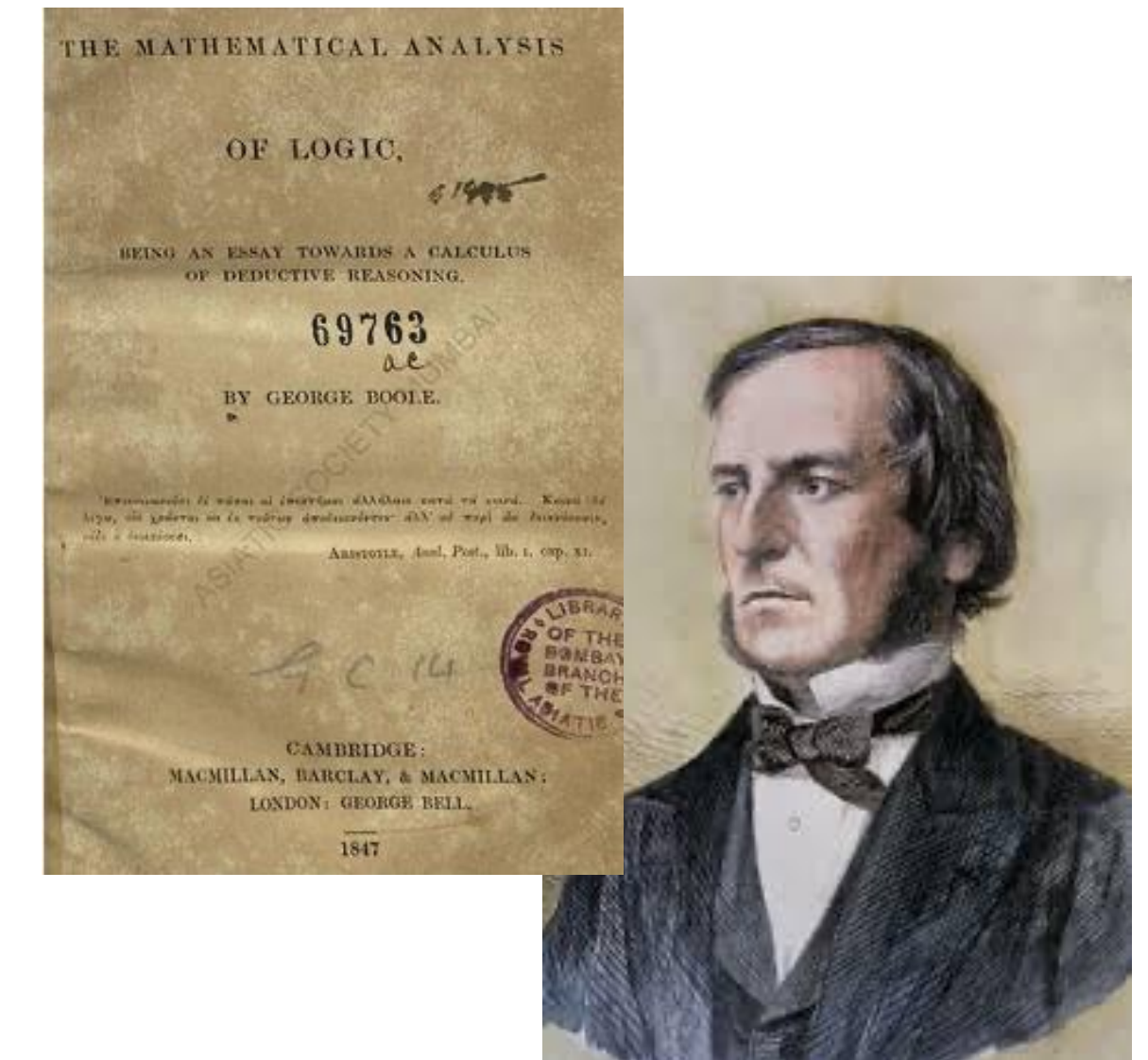
- **Has memory** or “history”: can “store” data values
- **Outputs** determined by **previous (history) and current values of inputs.**



In this session, we focus on **Combinational Logic**. **Next session**, more about **Sequential Logic**.

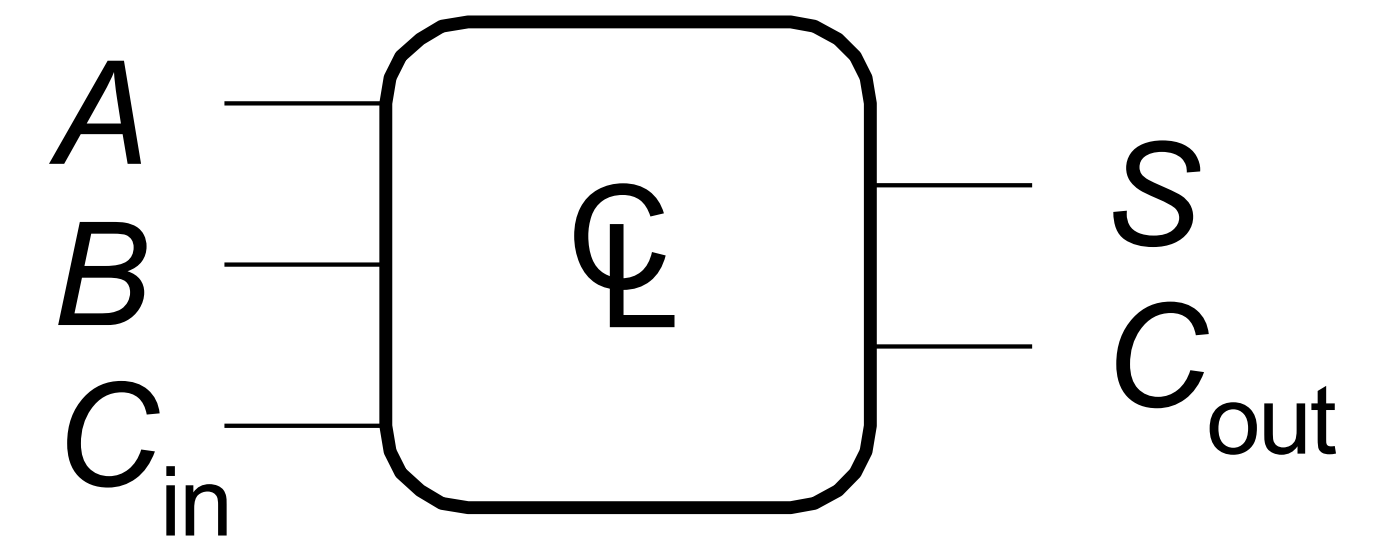
Boolean equations

- Defined using algebra of 1's and 0's.
- Base operations** OR, AND, NOT.
- George Boole**, "The Mathematical Analysis of Logic", 1847.



- Functional specification:**
- Defined with **outputs in terms of inputs**.
- Example (full 1-bit adder):**

$$S = F(A, B, C_{in})$$
$$C_{out} = F(A, B, C_{in})$$



$$S = A \oplus B \oplus C_{in}$$
$$C_{out} = AB + AC_{in} + BC_{in}$$

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Canonical Forms

Karnaugh Maps

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Important Definitions

Axiom:

- A **postulate** that is **taken to be true**.
- **Used to derive other statements**.

Theorem:

- **Proven to be true**.
- **Based on axioms or other statements**.

- **Complement:** variable with a bar over it

$\bar{A}, \bar{B}, \bar{C}$

- **Literal:** variable or its complement

$A, \bar{A}, B, \bar{B}, C, \bar{C}$

- **Implicant:** product of literals

ABC, AC, BC

- **Minterm:** product that includes all input variables

$ABC, \bar{A}\bar{B}C, \bar{A}B\bar{C}$

- **Maxterm:** sum that includes all input variables

$(A+B+C), (A+B+\bar{C}), (A+\bar{B}+C)$

Duality in Axioms and Theorems

- **Axioms and theorems** to **simplify Boolean equations**.
 - All the axioms come in “dual” form
 - Anything true for an expression also true for its dual
 - So any derivation you could make that is true, can be flipped into dual form, and it stays true
- Definition of **duality**:
 - A dual of a Boolean expression is derived by replacing
 - Every **AND** operation with an **OR** operation
 - Every **OR** operation with an **AND**
 - Every **constant 1** with **a constant 0**
 - Every **constant 0** with **a constant 1**
 - But don't change any of the literals or play with the complements!

Example $a \cdot (b + c) = (a \cdot b) + (a \cdot c)$

$$a + (b \cdot c) = (a + b) \cdot (a + c)$$

Boolean Axioms and Theorems

Number	Axiom	Dual	Name
A1	$B = 0 \text{ if } B \neq 1$	$B = 1 \text{ if } B \neq 0$	Binary Field
A2	$\overline{0} = 1$	$\overline{1} = 0$	NOT
A3	$0 \cdot 0 = 0$	$1 + 1 = 1$	AND/OR
A4	$1 \cdot 1 = 1$	$0 + 0 = 0$	AND/OR
A5	$0 \cdot 1 = 1 \cdot 0 = 0$	$1 + 0 = 0 + 1 = 1$	AND/OR

Number	Theorem	Dual	Name
T1	$B \cdot 1 = B$	$B + 0 = B$	Identity
T2	$B \cdot 0 = 0$	$B + 1 = 1$	Null Element
T3	$B \cdot B = B$	$B + B = B$	Idempotency
T4	$\overline{\overline{B}} = B$		Involution
T5	$B \cdot \overline{B} = 0$	$B + \overline{B} = 1$	Complements

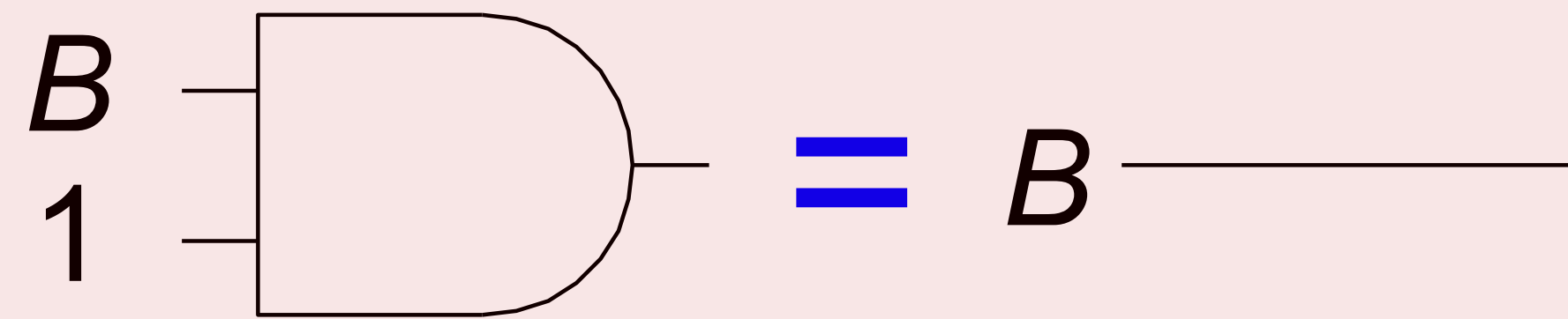
Dual
Replace:

- • with +
- 0 with 1

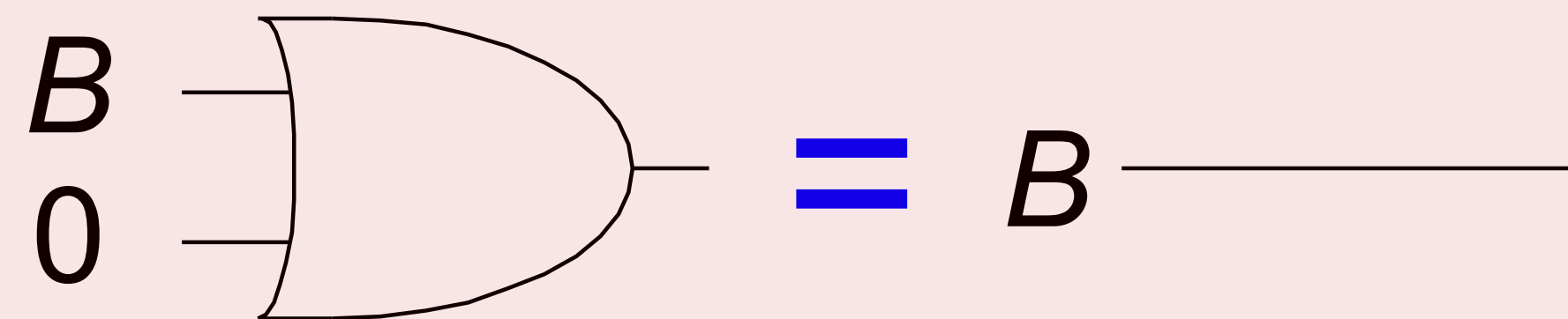
Graphical Representation

T1: Identity Theorem

- $B \cdot 1 = B$

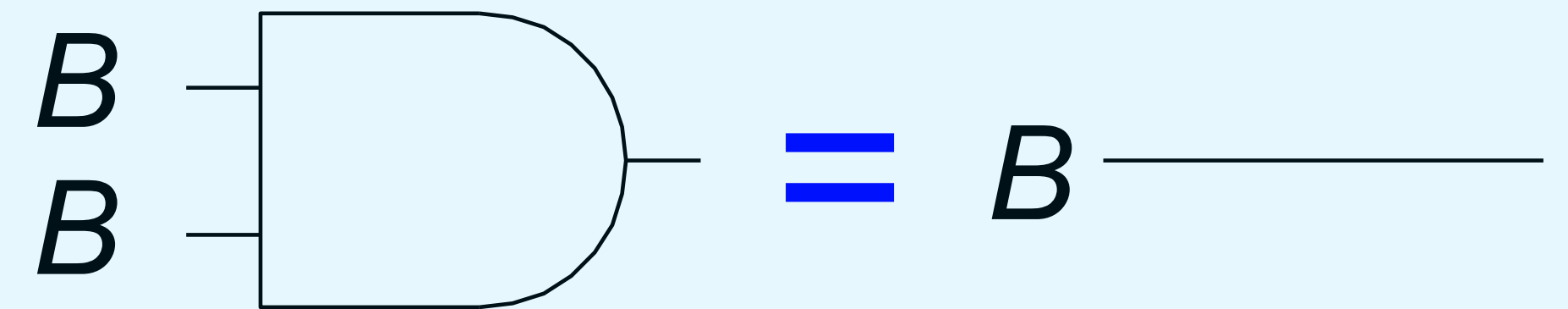


- $B + 0 = B$

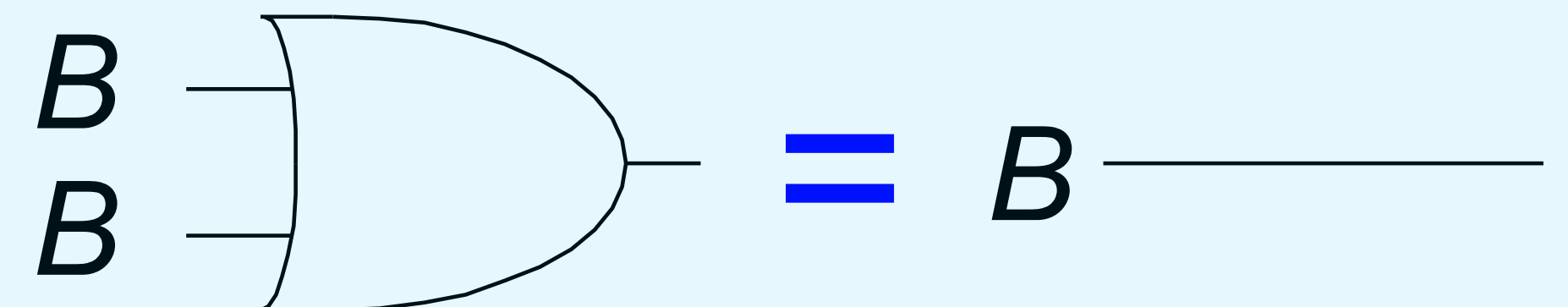


T3: Idempotency Theorem

- $B \cdot B = B$



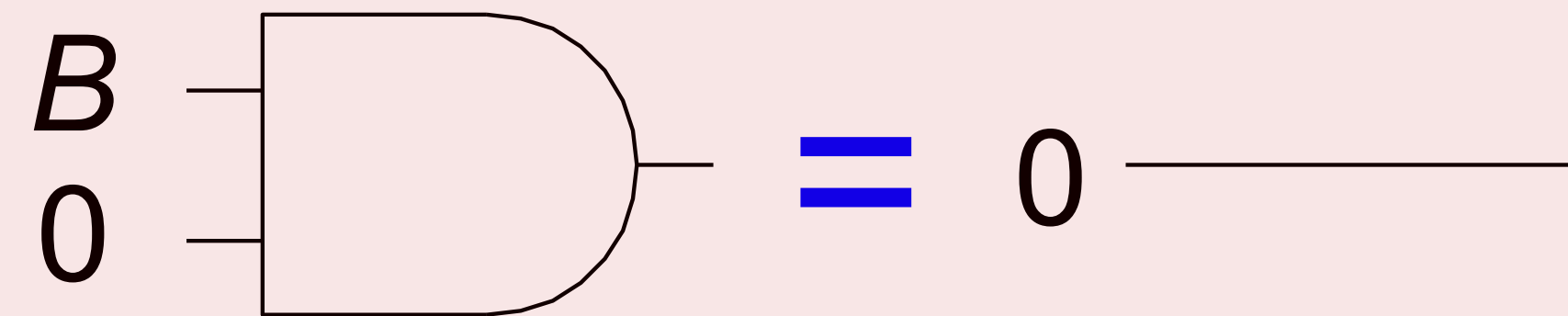
- $B + B = B$



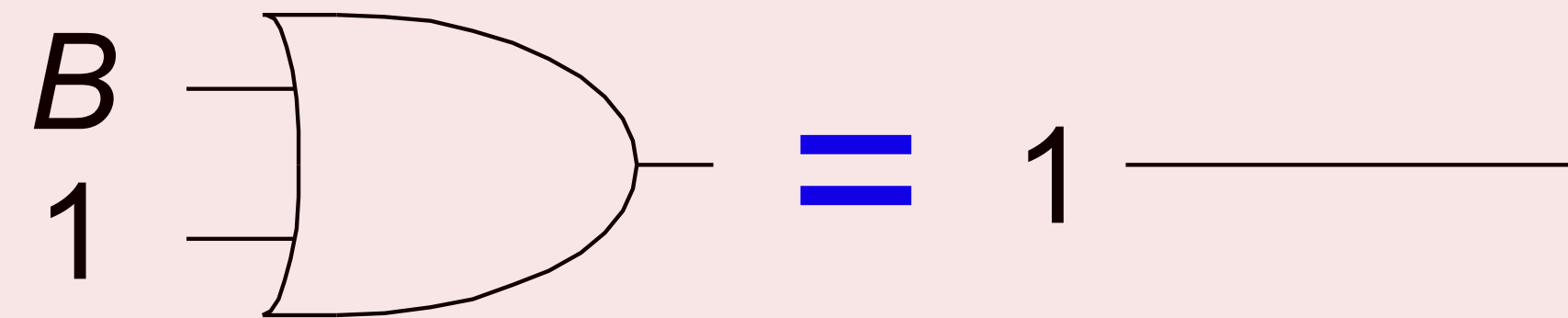
Graphical Representation

T2: Null Element Theorem

- $B \cdot 0 = 0$

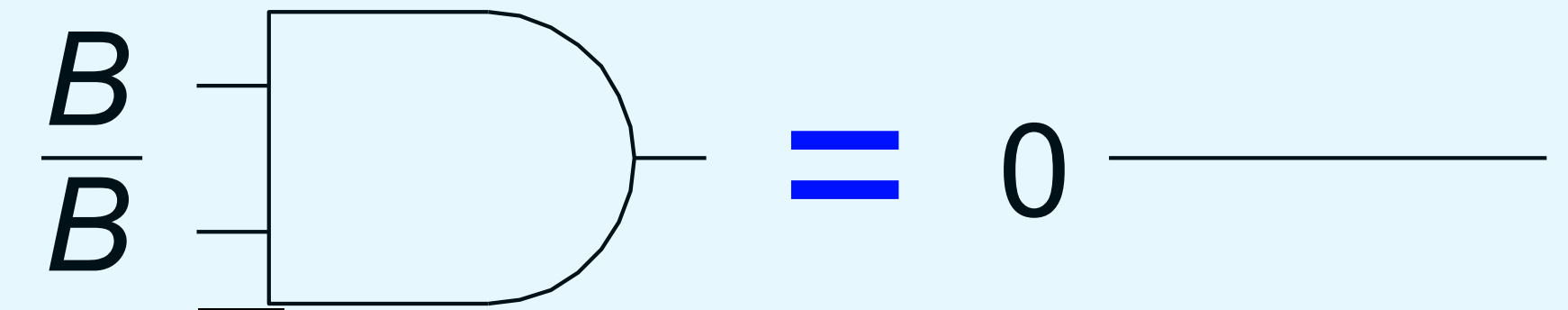


- $B + 1 = 1$

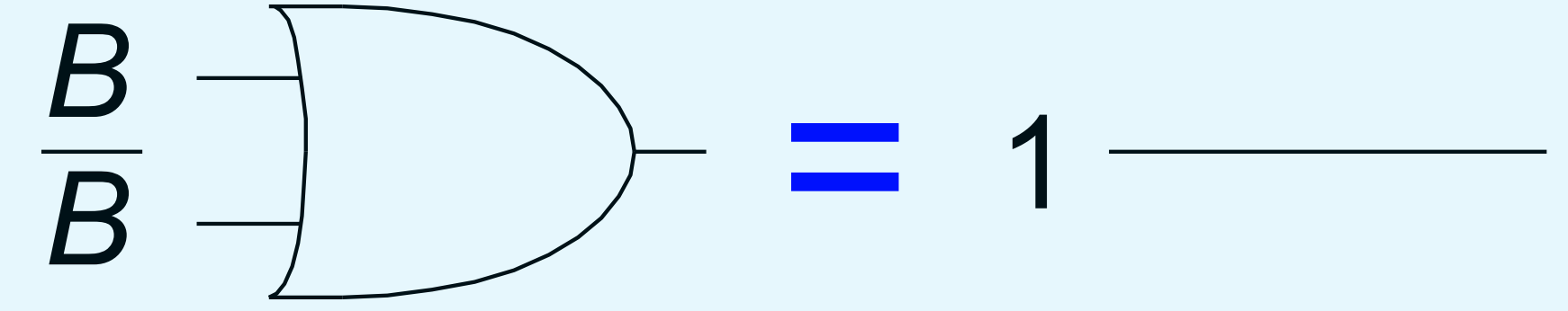


T5: Complement Theorem

- $B \cdot \overline{B} = 0$

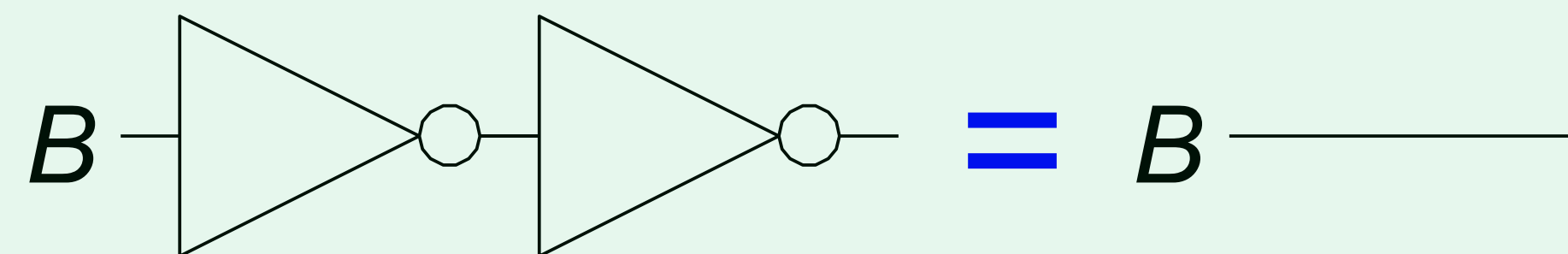


- $B + \overline{B} = 1$



T4: Identity Theorem

- $\overline{\overline{B}} = B$



Boolean Theorems of Multiple Variables

Number	Theorem	Name
T6	$B \cdot C = C \cdot B$	Commutativity
T7	$(B \cdot C) \cdot D = B \cdot (C \cdot D)$	Associativity
T8	$B \cdot (C + D) = (B \cdot C) + (B \cdot D)$	Distributivity
T9	$B \cdot (B + C) = B$	Covering
T10	$(B \cdot C) + (B \cdot \bar{C}) = B$	Combining
T11	$(B \cdot C) + (\bar{B} \cdot D) + (C \cdot D) = (B \cdot C) + (\bar{B} \cdot D)$	Consensus

If the **graphical observation is complex**, how can we prove these?

Commutativity: Proof by “perfect” induction

Number	Theorem	Name
T6	$B \cdot C = C \cdot B$	Commutativity

Commutativity: Proof by perfect induction

Number	Theorem	Name
T6	$B \cdot C = C \cdot B$	Commutativity

<i>B</i>	<i>C</i>	<i>BC</i>	<i>CB</i>
0	0	0	0
0	1	0	0
1	0	0	0
1	1	1	1

Covering: Proof by Perfect Induction

Number	Theorem	Name
T9	$B \cdot (B+C) = B$	Covering

Method 1: Perfect Induction

<i>B</i>	<i>C</i>	<i>(B+C)</i>	<i>B(B+C)</i>
0	0	0	0
0	1	1	0
1	0	1	1
1	1	1	1

Covering

Number	Theorem	Name
T9	$B \cdot (B + C) = B$	Covering

Method 2: Prove **true using other axioms and theorems.**

$$\begin{aligned} B \cdot (B + C) &= B \cdot B + B \cdot C \\ &= B + B \cdot C \\ &= B \cdot (1 + C) \\ &= B \cdot (1) \\ &= B \end{aligned}$$

T8: Distributivity
T3: Idempotency
T8: Distributivity
T2: Null element
T1: Identity

Boolean Theorems of Multiple Variables

#	Theorem	Dual	Name
T6	$B \cdot C = C \cdot B$	$B + C = C + B$	Commutativity
T7	$(B \cdot C) \cdot D = B \cdot (C \cdot D)$	$(B + C) + D = B + (C + D)$	Associativity
T8	$B \cdot (C + D) = (B \cdot C) + (B \cdot D)$	$B + (C \cdot D) = (B + C) (B + D)$	Distributivity
T9	$B \cdot (B + C) = B$	$B + (B \cdot C) = B$	Covering
T10	$(B \cdot C) + (B \cdot C) = B$	$(B + C) \cdot (B + C) = B$	Combining
T11	$(B \cdot C) + (B \cdot D) + (C \cdot D) = (B \cdot C) + (B \cdot D)$	$(B + C) \cdot (B + D) \cdot (C + D) = (B + C) \cdot (B + D)$	Consensus

Notice: T8 **differs** from traditional algebra: **OR (+) distributes over AND (•)**

Boolean Algebra: Proving Things

Proving theorems via other theorems of Boolean Algebra:

EX: Prove the theorem: $X \cdot Y + X \cdot \bar{Y} = X$

Distributive (T8)

Complement (T5)

Identity (T1)

EX2: Prove the theorem: $X + X \cdot Y = X$

Identity (T1)

Distributive (T8)

Identity (T1)

Identity (T1)

Boolean Algebra: Proving Things

Proving theorems via other theorems of Boolean Algebra:

EX: Prove the theorem: $X \cdot Y + X \cdot \bar{Y} = X$

$$X \cdot (Y + \bar{Y}) = X \quad \text{Distributive (T8)}$$

$$X \cdot 1 = X \quad \text{Complement (T5)}$$

$$X = X \quad \text{Identity (T1)}$$

EX2: Prove the theorem: $X + X \cdot Y = X$

$$X \cdot 1 + X \cdot Y = X \quad \text{Identity (T1)}$$

$$X \cdot (1 + Y) = X \quad \text{Distributive (T8)}$$

$$X \cdot 1 = X \quad \text{Identity (T1)}$$

$$X = X \quad \text{Identity (T1)}$$

Enabling Transformations

DeMorgan's Law:

$$\text{T12. } \overline{(X + Y + Z + \dots)} = \bar{X} \cdot \bar{Y} \cdot \bar{Z} \cdot \dots$$

$$\text{T12D. } \overline{(X \cdot Y \cdot Z \cdot \dots)} = \bar{X} + \bar{Y} + \bar{Z} + \dots$$

- Think of this **as a transformation**
 - Let's say we have:
$$F = A + B + C$$
 - Applying DeMorgan's Law (T12), gives us

$$F = \overline{\overline{(A + B + C)}} = \overline{(\bar{A} \cdot \bar{B} \cdot \bar{C})}$$

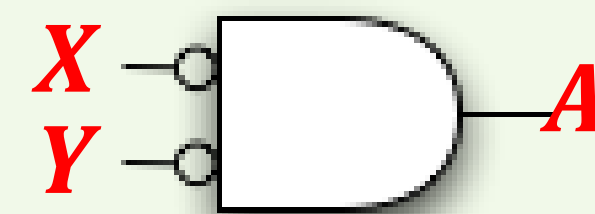
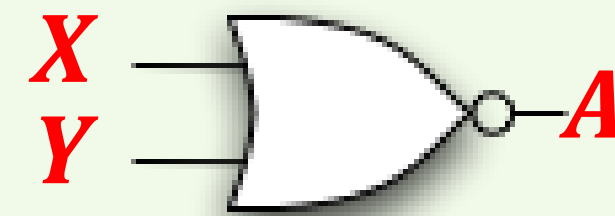
Evaluate: at least one of A, B, C is TRUE, then with A, B, C are all false

DeMorgan's Law

These are conversions between **different types of logic functions**
They can prove useful if you do not have **every type of gate**

$$A = \overline{(X + Y)} = \bar{X}\bar{Y}$$

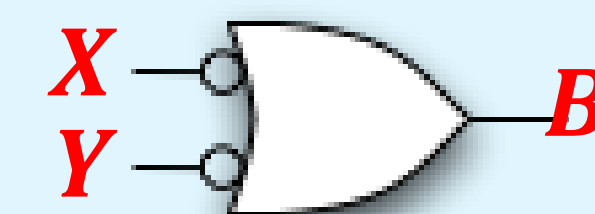
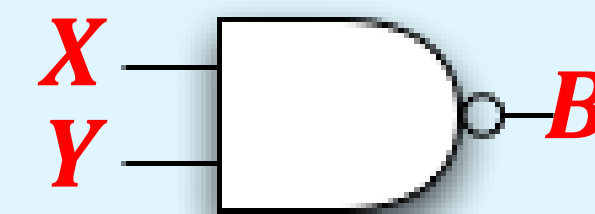
**NOR is equivalent to AND
with inputs complemented**



X	Y	$\overline{X + Y}$	$\bar{X}$	$\bar{Y}$	$\bar{X}\bar{Y}$
0	0	1	1	1	1
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	0

$$B = \overline{(XY)} = \bar{X} + \bar{Y}$$

**NAND is equivalent to OR
with inputs complemented**



X	Y	$\overline{XY}$	$\bar{X}$	$\bar{Y}$	$\bar{X} + \bar{Y}$
0	0	1	1	1	1
0	1	1	1	0	1
1	0	1	0	1	1
1	1	0	0	0	0

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Canonical Forms

Karnaugh Maps

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Two-Level Canonical (Standard) Forms

- **Truth table** is the unique **signature** of a Boolean *function* ...
 - But, it is an expensive representation
- **Boolean function** can have **many alternative Boolean expressions**
 - i.e., many alternative Boolean expressions (and gate realizations) may have the same truth table (and function)
- **Canonical** form: **standard form for a Boolean expression**
 - Provides a unique algebraic signature

Different Boolean expressions lead to different gate realizations

Sum-of-Products (SOP) Form

- **All equations** can be written in **SOP form**.
- Each row has a minterm.
- A **minterm** is a **product (AND)** of **literals**
- Each minterm is **TRUE** for that row (and **only that row**)
- Form function by **sum up (OR) minterms** where output is 1
- Thus, a sum (OR) **of products (AND terms)**

<i>A</i>	<i>B</i>	<i>Y</i>	minterm	minterm name
0	0	0	$\overline{A} \overline{B}$	m_0
0	1	1	$\overline{A} B$	m_1
1	0	0	$A \overline{B}$	m_2
1	1	1	$A B$	m_3

$$Y = F(A, B) = \overline{A}B + AB = \Sigma(1, 3)$$

Product-of-Sums (POS) Form

- All Boolean equations can be written in **POS form**
- Each row has a **maxterm**
- A maxterm is a **sum (OR) of literals**
- Each maxterm is **FALSE** for that row (and only that row)
- Form function by **multiply (AND) maxterms** where output is 0
- Thus, a product (AND) of sums (OR terms)

<i>A</i>	<i>B</i>	<i>Y</i>	maxterm	maxterm name
0	0	0	$A + B$	M_0
0	1	1	$A + \overline{B}$	M_1
1	0	0	$\overline{A} + B$	M_2
1	1	1	$\overline{A} + \overline{B}$	M_3

$Y = F(A, B) = (A + B)(\overline{A} + B) = \Pi(0, 2)$

Sum of products (SOP)

Also known as **disjunctive normal form** or **minterm expansion**

A	B	C	F	
0	0	0	0	
0	0	1	0	
0	1	0	0	
0	1	1	1	$\overline{A}BC$
1	0	0	1	$A\overline{B}\overline{C}$
1	0	1	1	$A\overline{B}C$
1	1	0	1	$AB\overline{C}$
1	1	1	1	ABC

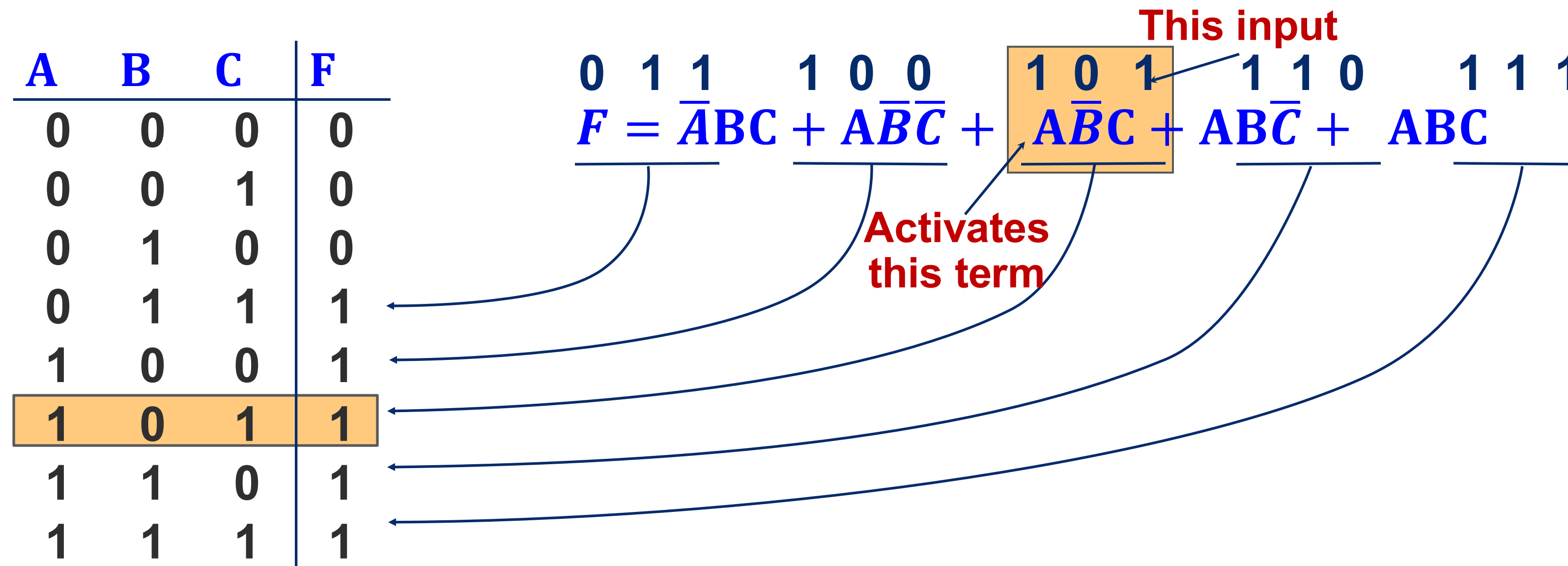
$F = \overline{A}BC + A\overline{B}\overline{C} + A\overline{B}C + AB\overline{C} + ABC$

- Each row in a truth table has a minterm
- A **minterm** is a product (AND) of literals
- Each **minterm is TRUE for that row** (and only that row)

All Boolean **equations can be written in SOP form**

Find all the input combinations (minterms) for which the output of the **function is TRUE**.

SOP Form: Why Does It Work?



- Only the shaded product term — $\bar{A}BC = 1 \cdot \bar{0} \cdot 1$ — will be 1
- **No other product terms** will “turn on” — they will all be 0
- So if inputs **A B C correspond to a product term** in expression,
 - We get $0 + 0 + \dots + 1 + \dots + 0 + 0 = 1$ for output
- If inputs **A B C do not correspond to any product term in expression**
 - We get $0 + 0 + \dots + 0 = 0$ for output

Aside: Notation for SOP

- Standard “shorthand” notation
 - **If we agree on the order** of the variables in the rows of truth table...
 - then **we can enumerate each row** with the decimal number that corresponds to the binary number created by the input pattern

A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

100 = decimal 4 so this is minterm #4, or m4

111 = decimal 7 so this is minterm #7, or m7

$$f = m3 + m4 + m5 + m6 + m7$$

$$= \sum m(3,4,5,6,7)$$

We can write this as a sum of products

Or, we can use a summation notation

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1	1	0	1
1	1	1	1

100 = decimal 4 so this is minterm #4, or m4

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$$f = m3 + m4 + m5 + m6 + m7$$
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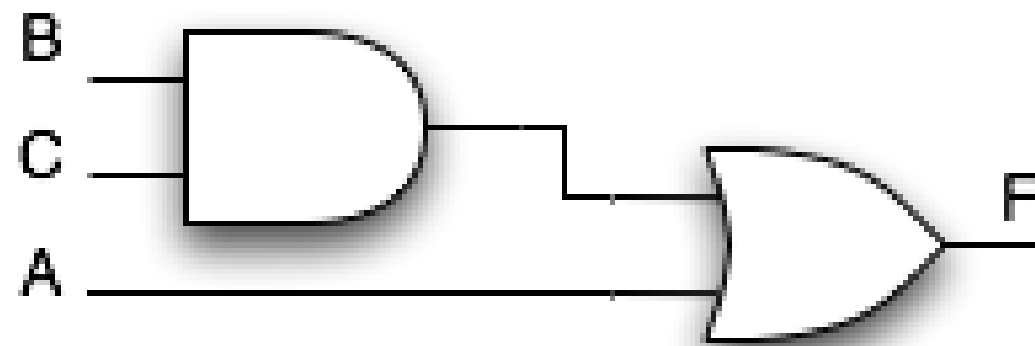
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Or, we can use a summation notation

Canonical SOP Forms

A	B	C	minterms
0	0	0	$\overline{A}\overline{B}\overline{C}$ = m0
0	0	1	$\overline{A}\overline{B}C$ = m1
0	1	0	$\overline{A}B\overline{C}$ = m2
0	1	1	$\overline{A}BC$ = m3
1	0	0	$A\overline{B}\overline{C}$ = m4
1	0	1	$A\overline{B}C$ = m5
1	1	0	$AB\overline{C}$ = m6
1	1	1	ABC = m7

Shorthand Notation for Minterms of 3 Variables



2-Level AND/OR Realization

F in canonical form:

$$F(A,B,C) = \sum m(3,4,5,6,7) \\ = m3 + m4 + m5 + m6 + m7$$

$$F = \overline{A}BC + A\overline{B}\overline{C} \\ + A\overline{B}C + AB\overline{C} + ABC$$

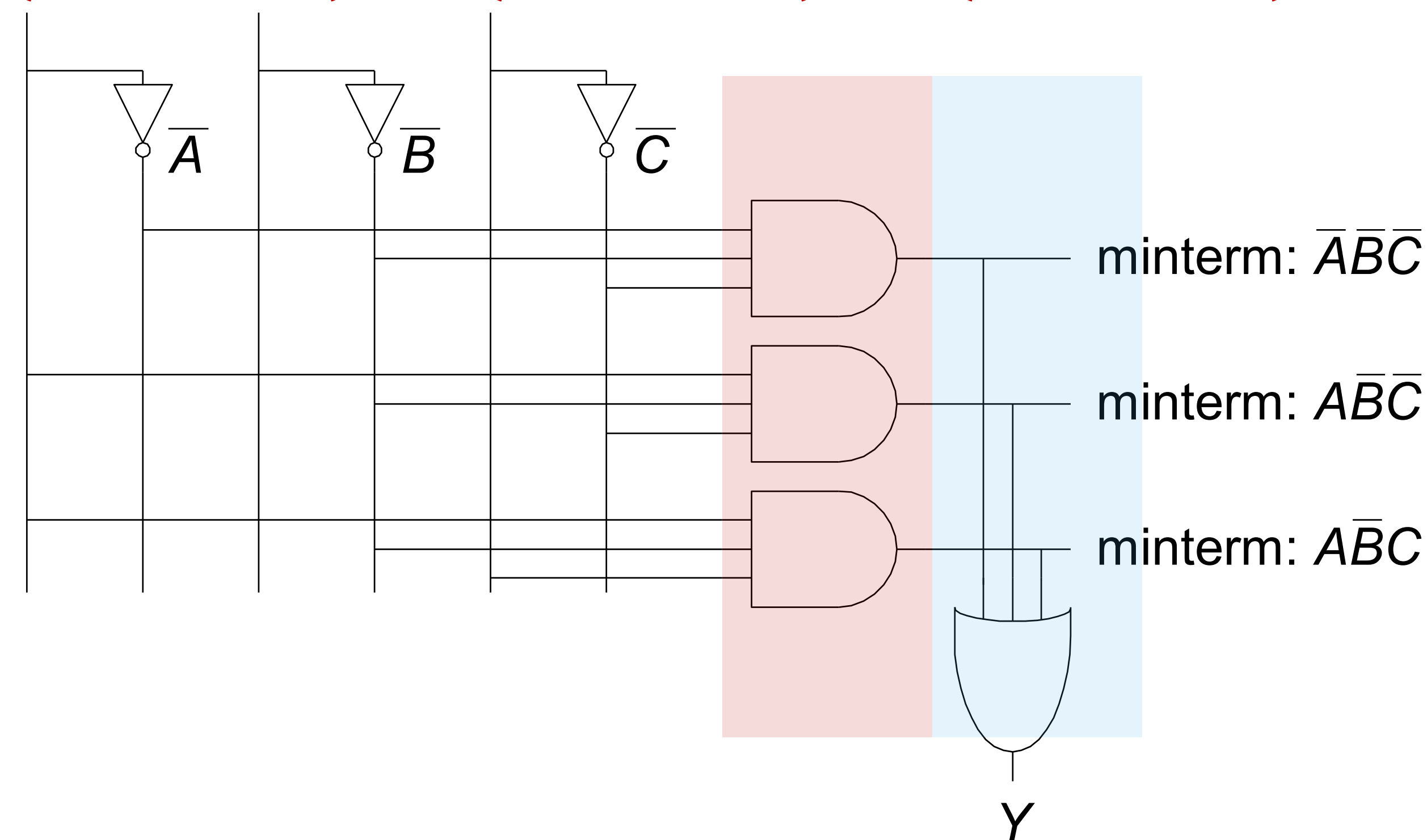
canonical form $\neq$ minimal form

$$F = A\overline{B}(C + \overline{C}) + \overline{A}BC + AB(C + \overline{C}) \\ = A\overline{B} + \overline{A}BC + AB \\ = A(\overline{B} + B) + \overline{A}BC \\ = A + \overline{A}BC \\ = A + BC$$

From Logic to Gates

- SOP (sum-of-products) leads to two-level logic

- Example: $Y = (\overline{A} \cdot \overline{B} \cdot \overline{C}) + (A \cdot \overline{B} \cdot \overline{C}) + (A \cdot \overline{B} \cdot C)$



Alternative Canonical Form: POS

DeMorgan of **SOP** of $\bar{F}$

A product of sums (POS)

products

$$F = (A + B + C)(A + B + \bar{C})(A + \bar{B} + C)$$

sums

Each sum term represents one of the “**zeros**” of the function

A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

0 0 0 0 0

$$F = (A + B + C)(A + B + \bar{C})(A + \bar{B} + C)$$

1	0	1
(A	(B	(C

This input

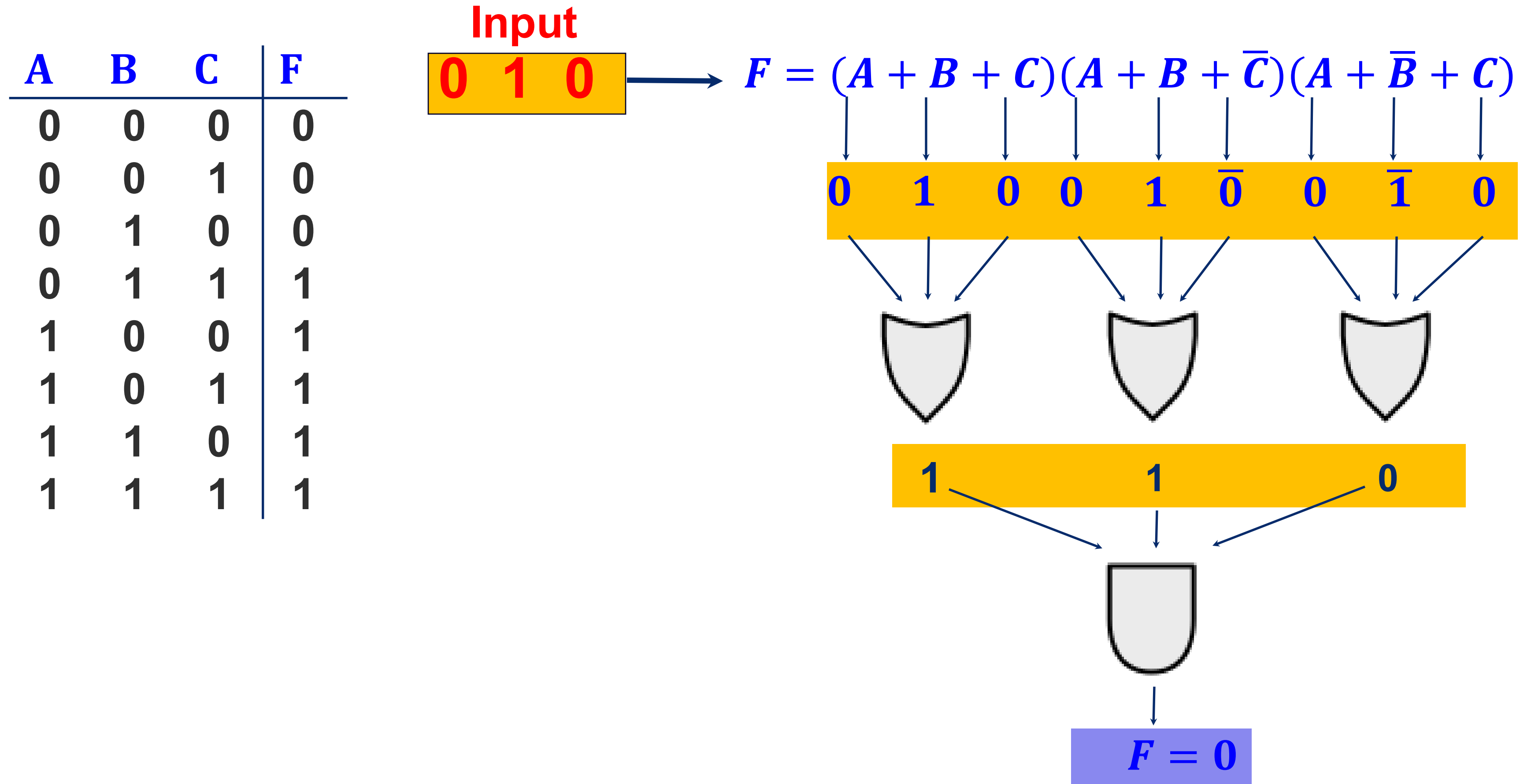
Activates this term

For the given input, only the shaded sum term will equal 0

$$A + \bar{B} + C = 0 + \bar{1} + 0$$

Anything ANDed with 0 is 0; Output F will be 0

Consider $A=0, B=1, C=0$



Only one of the products will be 0, anything ANDed with 0 is 0

Therefore, the output is $F = 0$

POS: How to Write It

A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

Maxterm form:

$$F = (A + B + C)(A + B + \bar{C})(A + \bar{B} + C)$$

$$A + \bar{B} + C$$

1. Find truth table rows where F is 0
2. 0 in input col -> true literal
3. 1 in input col -> complemented literal
4. OR the literals to get a Maxterm
5. AND together all the Maxterms

Or just remember, POS of F is the same as the DeMorgan of SOP of $\bar{F}$!!

Canonical POS Forms

Product of Sums / Conjunctive Normal Form / Maxterm Expansion

$$F = (A + B + C)(A + B + \bar{C})(A + \bar{B} + C)$$

$$\prod M(0, 1, 2)$$

A	B	C	Maxterms
0	0	0	$A + B + C = M0$
0	0	1	$A + B + \bar{C} = M1$
0	1	0	$A + \bar{B} + C = M2$
0	1	1	$A + \bar{B} + \bar{C} = M3$
1	0	0	$\bar{A} + B + C = M4$
1	0	1	$\bar{A} + B + \bar{C} = M5$
1	1	0	$\bar{A} + \bar{B} + C = M6$
1	1	1	$\bar{A} + \bar{B} + \bar{C} = M7$

**Maxterm shorthand notation
for a function of three variables**

A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

Note that you
form the
maxterms
around the
“zeros” of the
function

This is **not** the
complement of
the function!

Useful Conversions

- Minterm to Maxterm conversion:**
rewrite minterm shorthand using maxterm shorthand
replace minterm indices with the indices not already used
E.g., $F(A, B, C) = \sum m(3, 4, 5, 6, 7) = \prod M(0, 1, 2)$
- Maxterm to Minterm conversion:**
rewrite maxterm shorthand using minterm shorthand
replace maxterm indices with the indices not already used
E.g., $F(A, B, C) = \prod M(0, 1, 2) = \sum m(3, 4, 5, 6, 7)$

- ### 3. Expansion of $\mathbf{F}$ to expansion of $\overline{F}$:

$$\begin{array}{ccc} \text{E. g., } F(A, B, C) = \sum m(3, 4, 5, 6, 7) & \longrightarrow & \bar{F}(A, B, C) = \sum m(0, 1, 2) \\ = \prod M(0, 1, 2) & \longrightarrow & = \prod M(3, 4, 5, 6, 7) \end{array}$$

4. **Minterm expansion of F to Maxterm expansion of $\bar{F}$:**
rewrite in Maxterm form, using the same indices as F

E. g., $F(A, B, C) = \sum m(3, 4, 5, 6, 7)$ $\longrightarrow$ $\bar{F}(A, B, C) = \prod M(3, 4, 5, 6, 7)$
 $= \prod M(0, 1, 2)$ $\longrightarrow$ $= \sum m(0, 1, 2)$

Outline

Introduction

Definitions, axioms and theorems

Canonical Forms

Karnaugh Maps

Conclusion

Karnaugh Maps (K-Maps)

- Boolean expressions **can be minimized** by **combining terms**
- K-maps** minimize equations graphically
- $PA + PA = P$

A	B	C	Y
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	0

Y C \ AB		00	01	11	10
		0	1	1	1
0	0	1	0	0	0
	1	1	0	0	0

Y C \ AB		00	01	11	10
		0	1	1	1
0	0	$\bar{A}\bar{B}\bar{C}$	$\bar{A}B\bar{C}$	$AB\bar{C}$	$A\bar{B}\bar{C}$
	1	$\bar{A}\bar{B}C$	$\bar{A}BC$	ABC	$A\bar{B}C$

K-Map

- Circle 1's in **adjacent squares**
- In Boolean expression, **include only literals whose true and complement form are *not* in the circle**

A	B	C	Y
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	0

		AB			
Y	C	00	01	11	10
	0	1	0	0	0
	1	1	0	0	0

$$Y = \overline{A}\overline{B}$$

K-Map Definitions

- **Complement:** variable with a bar over it

A, B, C

- **Literal:** variable or its complement

$A, \bar{A}, B, \bar{B}, C, \bar{C}$

- **Implicant:** product of literals

$ABC, \bar{A}\bar{C}, BC$

- **Prime implicant:** implicant corresponding to the largest circle in a K-map

K-Map Rules

- . **Every 1 must be circled** at least once
- . Each **circle must span a power** of 2 (i.e. 1, 2, 4) squares in each direction
- . Each **circle must be as large as possible**
- . A **circle may wrap around** the edges
- . A **“don't care” (X)** is circled only if it helps minimize the equation

4-Input K-Map

<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>Y</i>
0	0	0	0	1
0	0	0	1	0
0	0	1	0	1
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	0
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

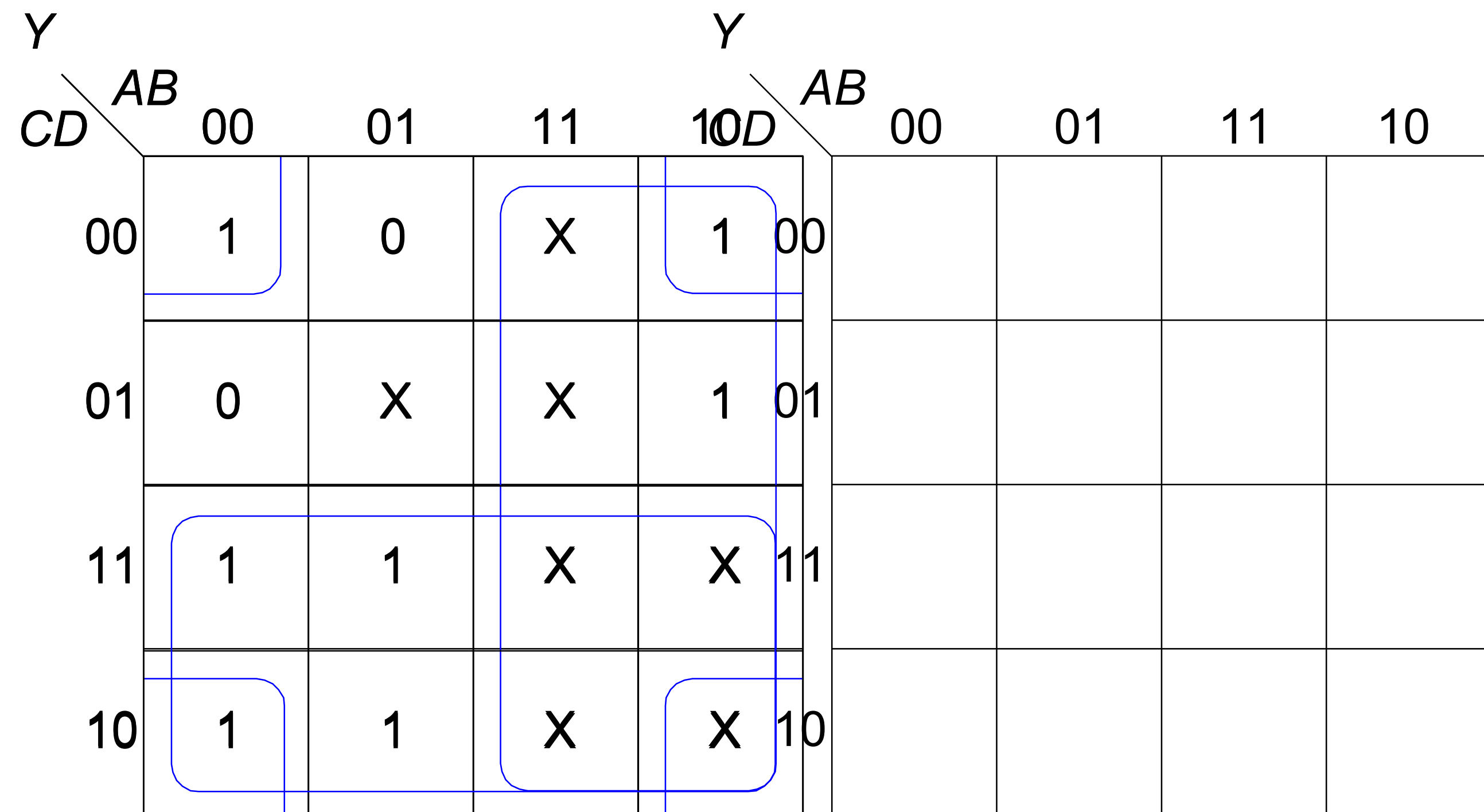
<i>Y</i> <i>CD</i> \ <i>AB</i>	00	01	11	10
00	1	0	0	1
01	0	1	0	1
11	1	1	0	0
10	1	1	0	1

<i>Y</i> <i>CD</i> \ <i>AB</i>	00	01	11	10
00	1	0	0	1
01	0	1	0	1
11	1	1	0	0
10	1	1	0	1

$$Y = \bar{A}\bar{C} + \bar{A}BD + A\bar{B}\bar{C} + \bar{B}\bar{D}$$

K-Maps with Don't Cares

<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>Y</i>
0	0	0	0	1
0	0	0	1	0
0	0	1	0	1
0	0	1	1	1
0	1	0	0	0
0	1	0	1	X
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1
1	0	1	0	X
1	0	1	1	X
1	1	0	0	X
1	1	0	1	X
1	1	1	0	X
1	1	1	1	X



$$Y = A + \overline{B}\overline{D} + C$$

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- **We introduced fundamentals concepts** about logic circuits.
- **Understanding mathematical definitions** allows to simplify Boolean equations.
- **Each expression represents an specific equivalent circuit.**
- We conclude that complex combinational circuits can be defined with Boolean equations **(and reduces them with analysis).**

2. Combinational Logic

Computer Architecture